MadhuTwin

A digital twin for Type 2 Diabetes in India that warns the doctor about a glucose spike two hours before it happens, and shows what would prevent it.

Madhu (मधु) = honey; madhumeha is the Ayurvedic name for diabetes.

Team Vantage
Management Development Institute, Gurgaon
Happiest Health Digital Twin Challenge 2026 · Phase 1 submission

India has 101 million people with diabetes

and their glucose is managed in 3-monthly snapshots, not in the hours when it actually goes wrong

101 M

adults with diabetes (11.4%)

136 M

more with prediabetes

~70 g+

carbs in a typical Indian lunch; rice, roti and sweet chai drive steep post-meal spikes

What a doctor sees today

- An HbA1c every 3 months: an average that hides daily spikes
- A CGM report after the fact, if the patient has a sensor
- Generic diet advice that ignores how this person responds
- No early warning and no way to test a change before prescribing it

What MadhuTwin adds

- A virtual replica of each patient, updated every 5 minutes
- A 2-hour warning before glucose crosses 180 mg/dL
- Plain-language reasons: meal habit, sleep, stress, activity
- A what-if simulator: rice → millet, smaller portion, walk after eating

Source: ICMR-INDIAB-17, Lancet Diabetes & Endocrinology, 2023.

Use case: predict post-meal spikes 2 hours ahead

Condition: Type 2 Diabetes · Adverse event: an excursion ≥180 mg/dL lasting ≥15 minutes (International Consensus on Time in Range)

Static EHR

Demographics, diagnoses, meds, labs (HbA1c, HOMA-IR, HOMA-B, eGFR), 12-month trends

→

Wearables

CGM every 5 min, heart rate, HRV, steps, sleep stages, food log

→

Layer 1 · Twin

Personal insulin sensitivity, beta-cell response, basal glucose; hidden state every 5 min; 2-h projection

→

Layer 2 · AI

LightGBM risk ± confidence, time to event, forecasts; reasons in plain words

→

Doctor console

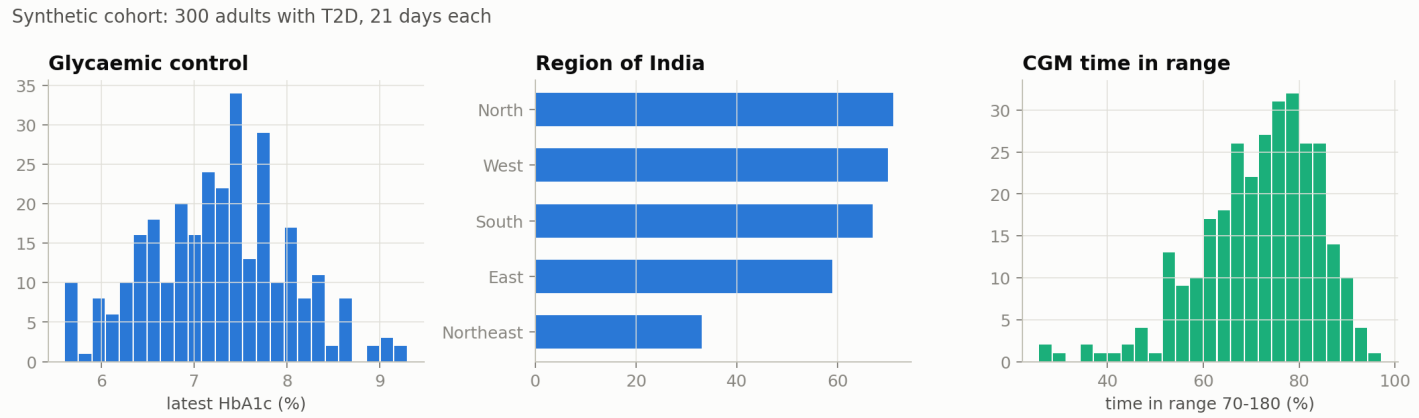
Who → when → why → what to do → what if I intervene?

Why a hybrid twin, not just a predictor

A pure ML model can say what will probably happen. A mechanistic twin can also answer what would happen if the patient changed something, and it can be personalised from day 0 using the health record. MadhuTwin uses both: physiology for personalisation and counterfactuals, ML for accuracy on messy real-life data.

Data: 300 India-context synthetic patients

EHRs in the Synthea CSV schema; wearables from a physiology simulator of Indian days · plus a separate retrospective check on 265 adults from open cohorts (USA + Spain)



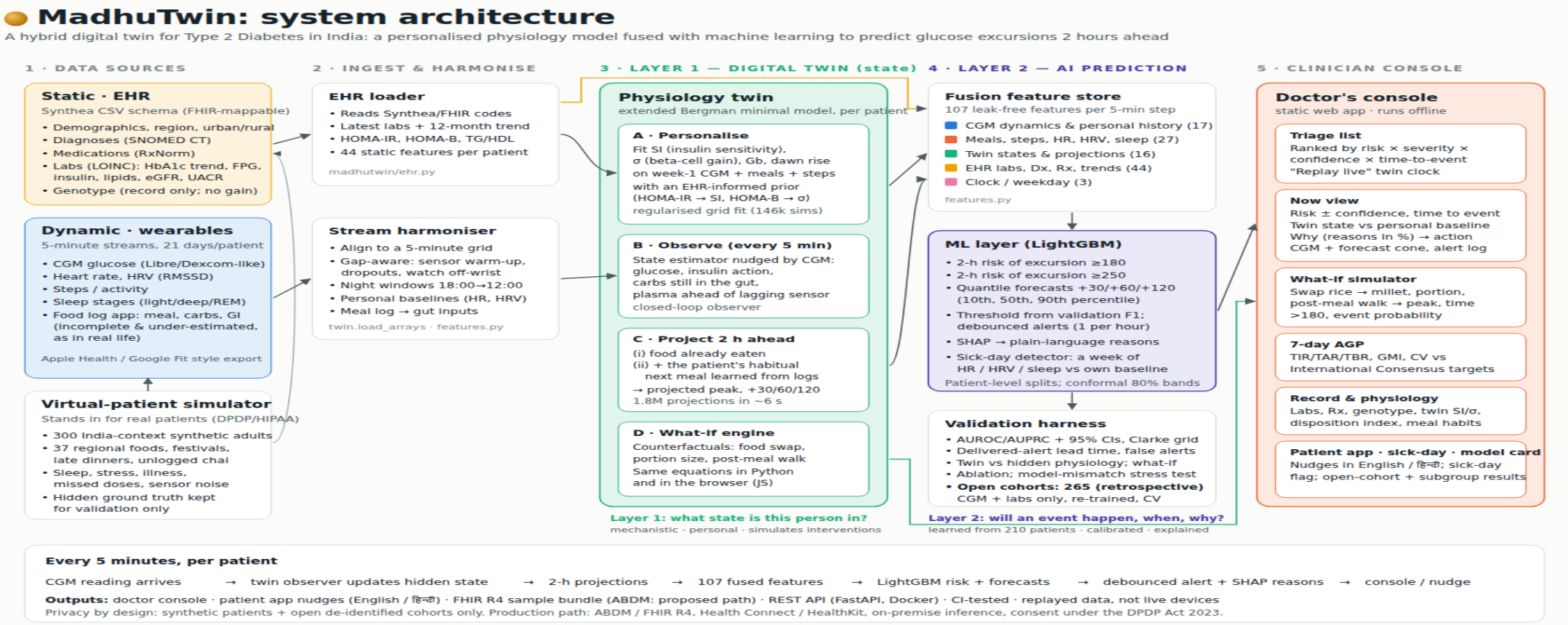
Stream	Resolution	Size
CGM glucose	5 min	1.77 M readings
HR · HRV · steps · sleep	5 min	6,048 per patient per signal
Food log	event	15,684 meals
EHR (Synthea schema)	static	labs, SNOMED CT, LOINC, RxNorm

What makes it realistic

- 37 regional foods with carbs and GI: idli, poha, rajma chawal, fish-rice, millets
- Late dinners, sugary chai (often unlogged), festival thalis, eating out
- Sleep with 90-min cycles, stressful days, illness, missed doses
- CGM lag, noise, drift and dropouts; watch off-wrist while charging
- Food log completed only ~56% of the time, carbs under-estimated
- Hidden 'true' physiology kept aside to validate the twin

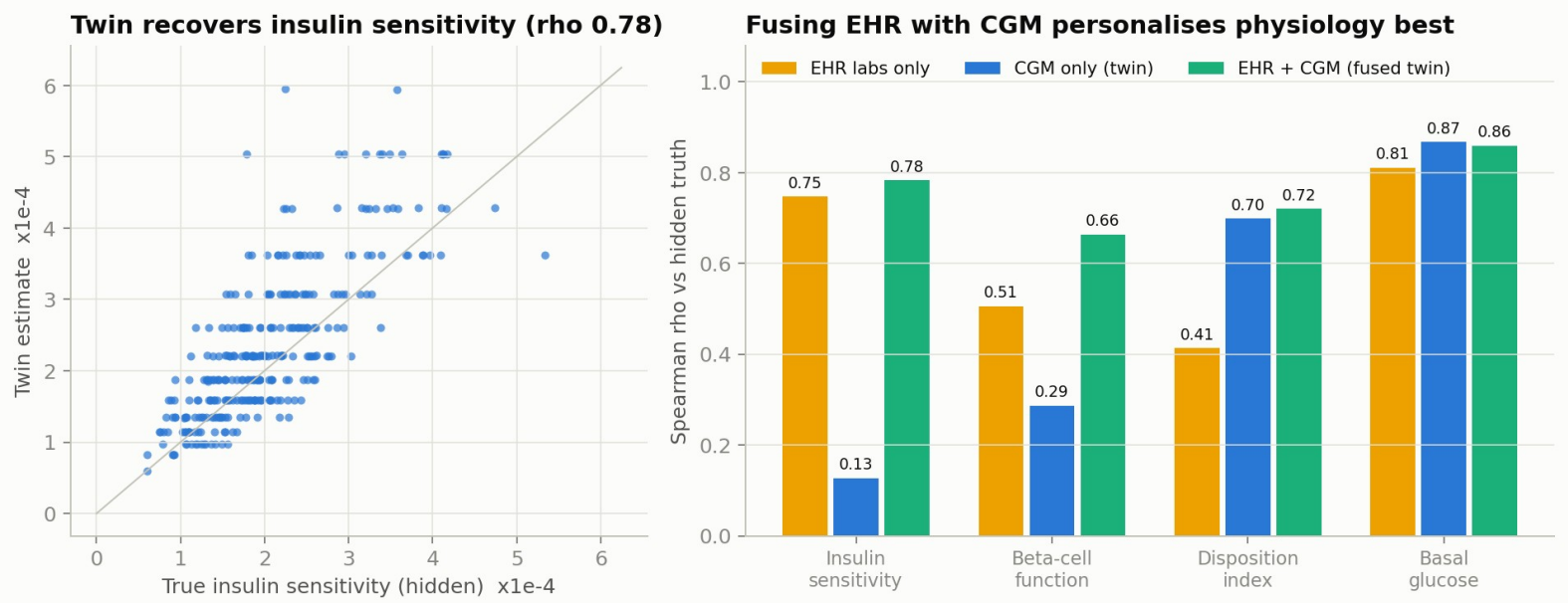
Architecture

Layer 1 keeps a live physiological state per patient; Layer 2 predicts from it. Every arrow runs every 5 minutes.



The twin learns each patient's physiology

Personalised from an EHR-informed prior + week-1 CGM (regularised fit), then checked against the simulator's hidden truth



Fusion matters here

CGM alone cannot separate insulin resistance from weak insulin secretion (rho 0.13). Labs alone are noisy. Together, the twin recovers insulin sensitivity at rho 0.78 and beta-cell function at 0.66.

A · Personalise

Fit SI, β-cell gain, basal glucose, dawn rise

B · Observe

Track carbs in gut, insulin action, true plasma glucose

C · Project

Simulate 2 h: food eaten + usual next meal

D · What-if

Re-simulate with a food swap, smaller portion or walk

A digital twin, not just another risk model

How the EHR and wearables are fused into one live patient state

	Ordinary AI risk model	MadhuTwin
Answers	“Will glucose go high?”	“What state is this person in, what happens next, what if we act?”
Per-patient model	No; shared weights	Yes: SI, β-cell gain, basal glucose, dawn rise
New patient	Needs history	Starts from the EHR on day 0; matures over week 1
Interventions	Cannot simulate	Food swap, portion, walk: simulated estimates
Drift	Retrain	Nightly re-personalisation

How fusion works

- EHR → informed prior for a regularised fit of the twin's physiology (HOMA-IR → insulin sensitivity, HOMA-B → β-cell gain)
- Wearables → the twin's observer every 5 min (CGM), its inputs (meals, steps) and personal baselines (HR, HRV, sleep)
- Twin state + raw streams + EHR → Layer 2 AI

Patient state vector	Updated
Glucose (plasma, ahead of sensor)	5 min
Carbs still in the gut	5 min
Insulin action	5 min
Activity, HR vs own baseline	5 min
Sleep debt, HRV vs baseline	nightly
Insulin sensitivity (× baseline)	nightly
Basal glucose	nightly
β-cell gain, dawn rise	weekly
HbA1c, eGFR, medicines	new result
Habitual next meal	weekly

It warns before the spike

45 held-out patients the model never saw · days 8–21 · a decision every 5 minutes

0.912

AUROC, ≥180 mg/dL within 2 h (95% CI 0.905–0.918)

75%

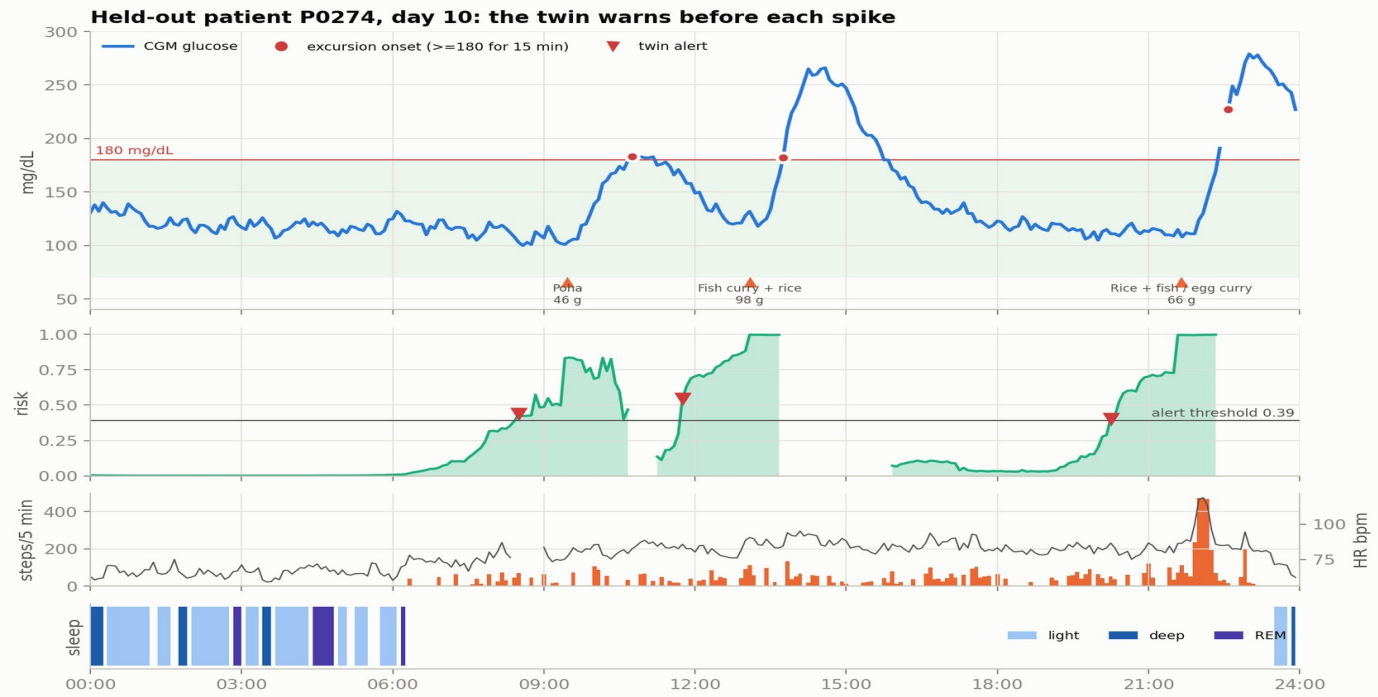
of 2,001 spikes warned ≥30 min ahead by delivered alerts (CI 70%–79%)

75 min

median warning before onset

2.3

false alerts per patient-day (threshold 0.40)

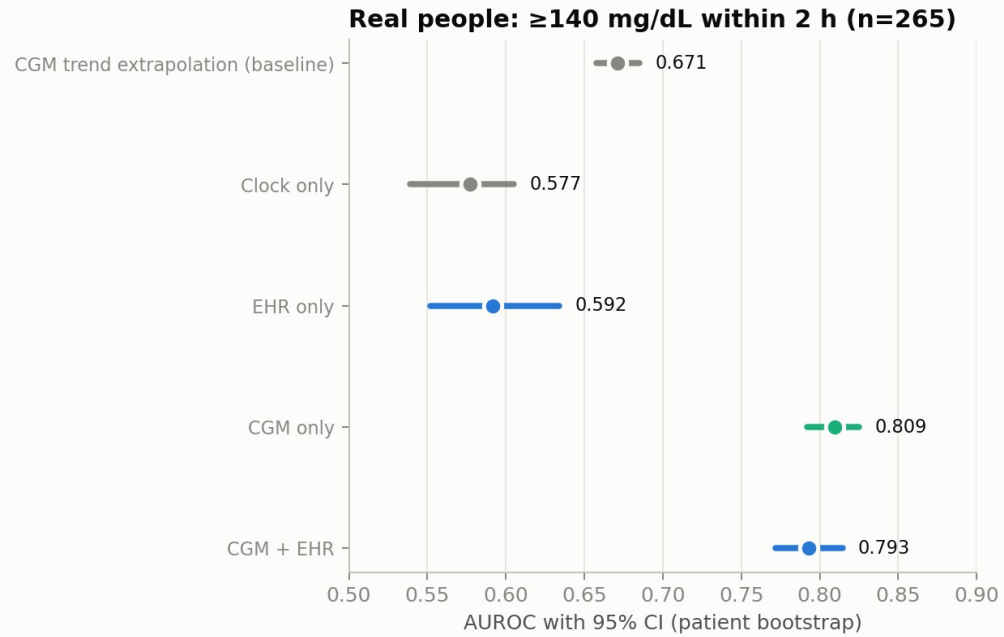


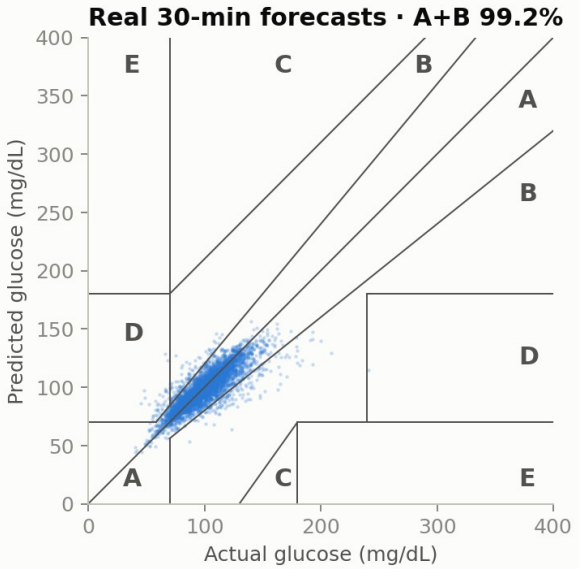
Operating threshold 0.40: why

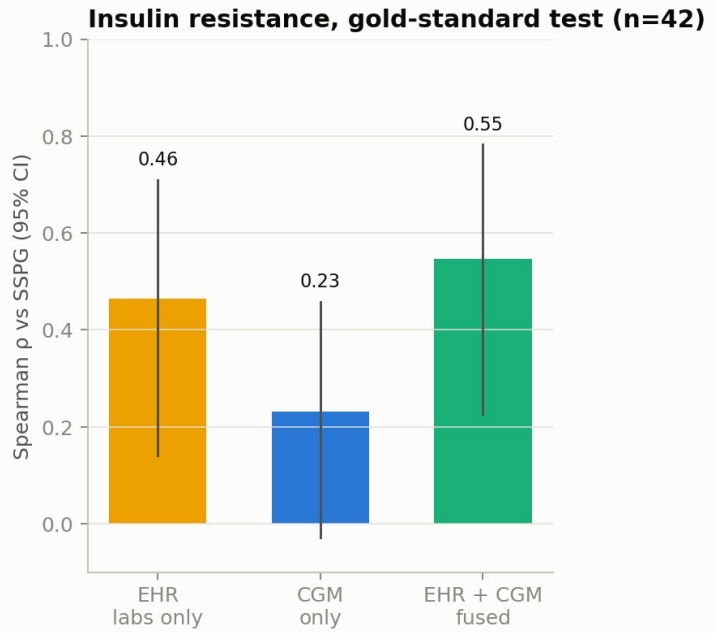
- Chosen on validation patients (best F1), never on test
- Debounced to ≤1 alert per hour. “We optimise for useful alerts, not maximum sensitivity.”
- A busy clinic can pick 0.70: 0.8 false alerts/day, still 63% of spikes ≥30 min early
- When: onset time error 17 min; 2-h peak error 18 mg/dL
- Calibrated: ECE 0.009, Brier 0.105; confidence from a 5-model ensemble

Retrospective check on open cohorts

265 adults (183 prediabetes, 72 normoglycaemic, 10 T2D) · CGM + labs only · re-trained, patient-level CV · target ≥140 mg/dL







AUROC 0.809 / 0.793

CGM-only / CGM + EHR, ≥140 mg/dL in 2 h (trend baseline 0.671); not the full system

99.2% in Clarke zones A+B

30-min forecasts: 6.7 mg/dL error vs 7.9 for 'no change'

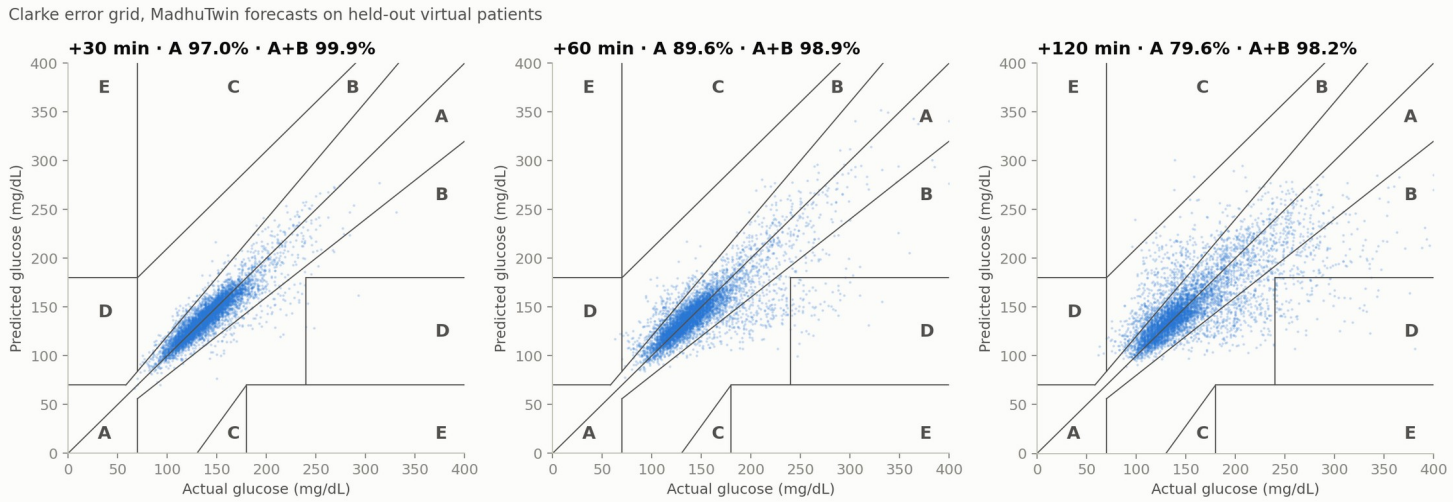
Fusion wins on physiology

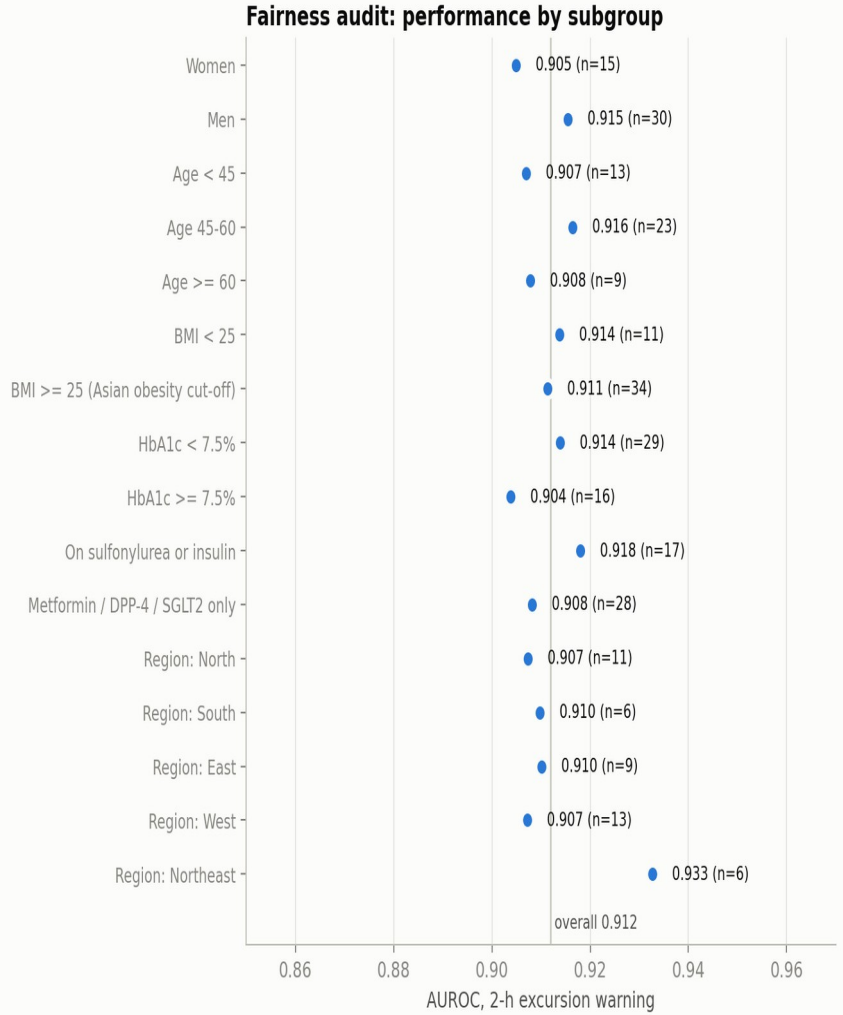
Insulin resistance (gold-standard SSPG): labs + CGM ρ 0.55 vs labs 0.46, CGM 0.23 (n=42); the alarm itself is CGM-driven

Not external validation of the synthetic-trained twin: the pipeline was re-trained and cross-validated within these cohorts (no person in both train and test). Only 10 people have T2D; meals, wearables, the full twin and Indian patients are untested on real data.

Evaluation details

Held-out synthetic evaluation · Clarke error grid · patient-level 95% CIs · empirical forecast bands · exploratory subgroups



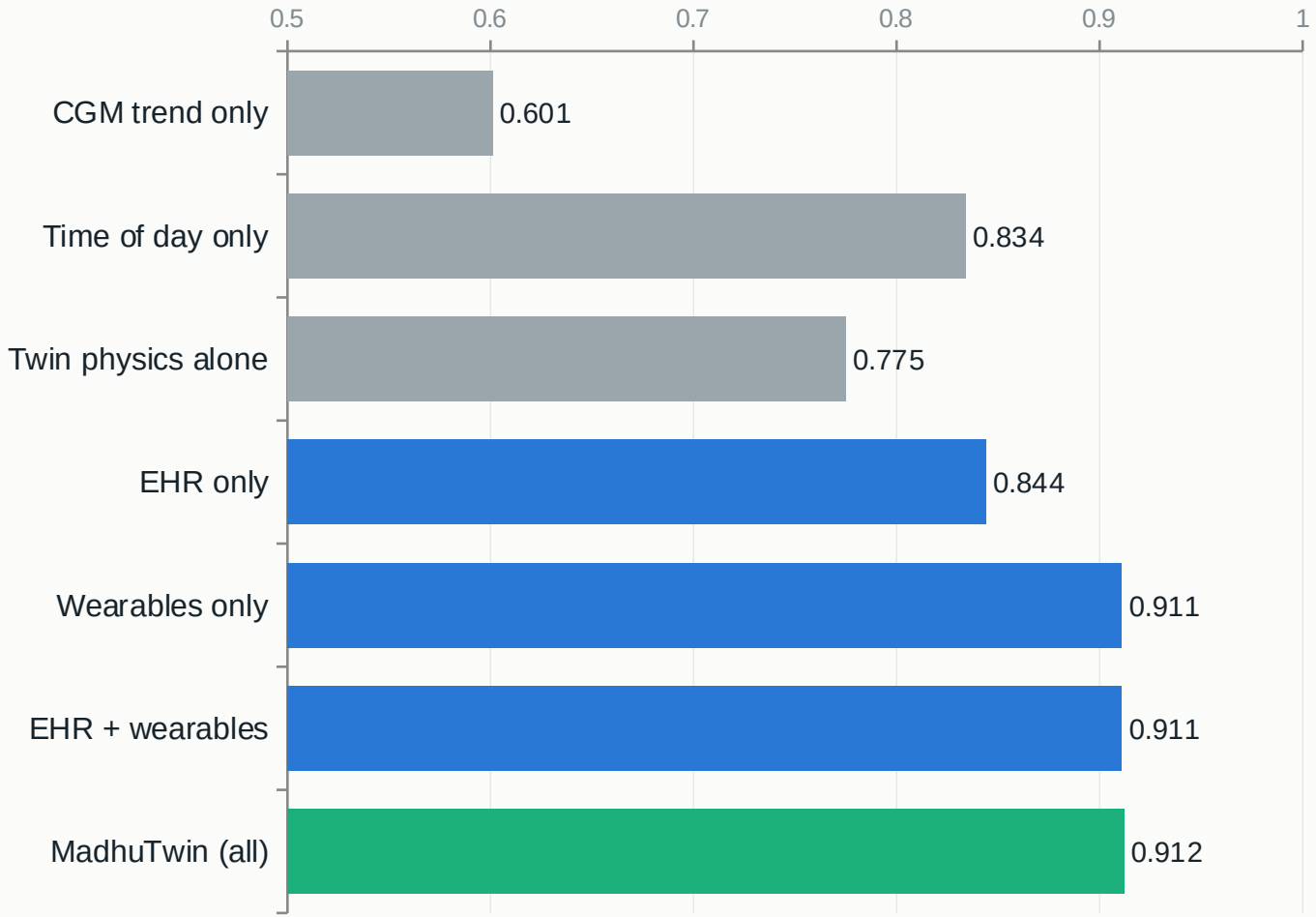


How the numbers were checked

- Forecasts: 99.9% / 98.9% / 98.1% in Clarke zones A+B at 30 / 60 / 120 min; zone D+E (dangerous) 0.13% at 30 min
- Empirical coverage of split-conformal 80% bands: 79% at 60 min (raw quantiles 74%); observed on this test set, not guaranteed
- Subgroups (exploratory, small): AUROC 0.903–0.932 across sex, age, BMI, HbA1c, medication and region
- Every headline number carries a 95% CI from resampling patients

What each data stream contributes

AUROC on held-out patients; we report what we found, including where fusion did not help

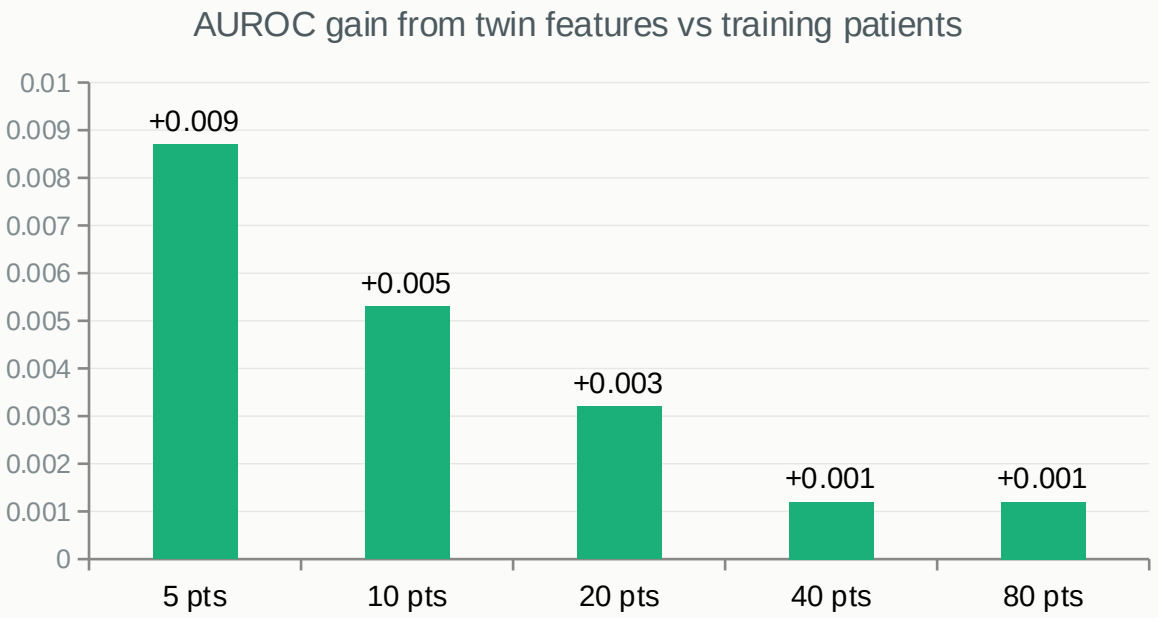


What we learned

- Once a patient has a week of CGM, their own glucose history carries most of the signal
- EHR alone is informative (0.844) but adds little on top of wearables for the 2-h alarm
- Where the EHR matters: personalising physiology (insulin sensitivity rho 0.13 → 0.78) and the what-if simulator
- Genotype added nothing (AUROC 0.9123 with vs 0.9117 without), so the risk model leaves it out
- The twin's value is in low data and counterfactuals: next slide

What the twin measurably adds

Same LightGBM, same data, with vs without twin features · held-out patients



0.13→0.78

insulin-sensitivity ρ vs truth: CGM-only → fused twin

0.45→0.72

disposition index ρ: day 0 (EHR only) → day 7

0.93

ρ predicted vs true rice→millet effect (ML alone: n/a)

×0.85

insulin sensitivity tracked nightly during illness (×0.99 when well)

Honest reading

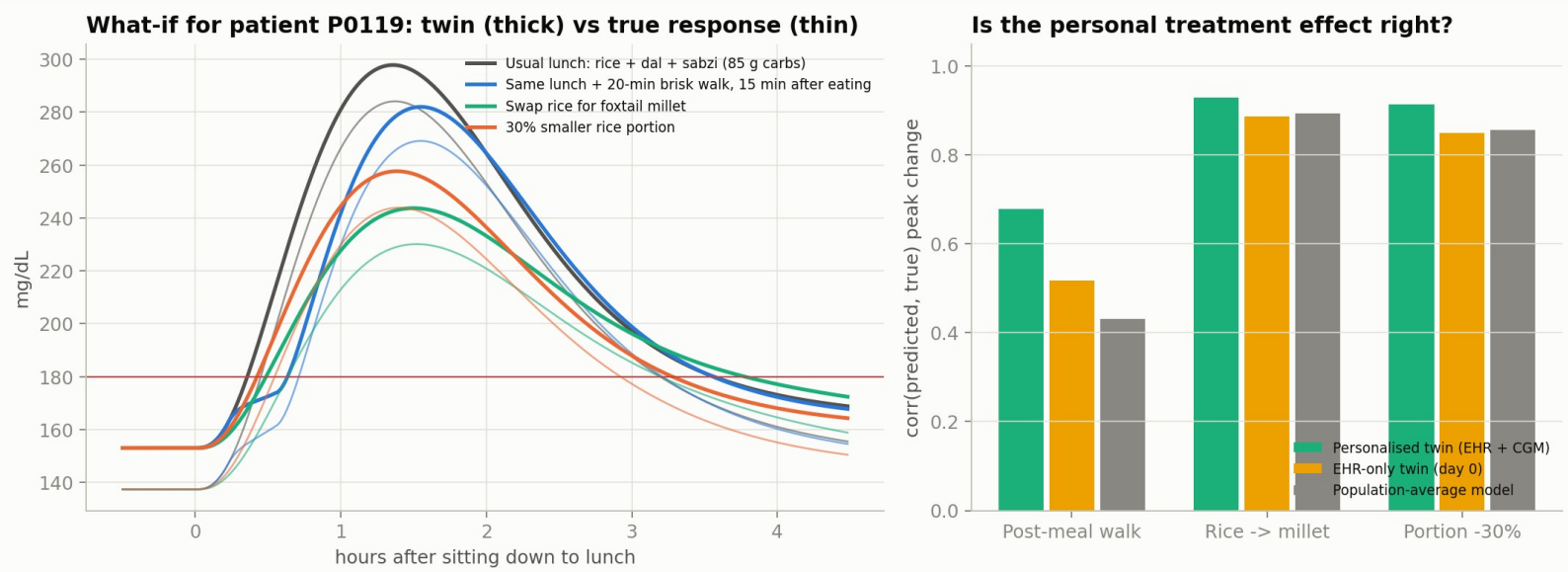
With all 210 training patients the gain is within noise (+0.0007, 95% CI -0.0004 to +0.0020); false alerts at equal sensitivity 2.23 → 2.30/day. The twin helps most where data are scarce: a new clinic or a new patient.

Twin maturity

- Day 0: personalised from the EHR alone, no CGM history needed
- Days 1–7: physiology sharpens (DI ρ 0.45 → 0.56 → 0.69 → 0.69 → 0.72 → 0.72)
- After week 1: nightly re-fit tracks illness, stress and missed doses

What-if simulator vs synthetic ground truth

Each intervention was re-run on the hidden 'true' virtual patient · real-patient effects remain unvalidated



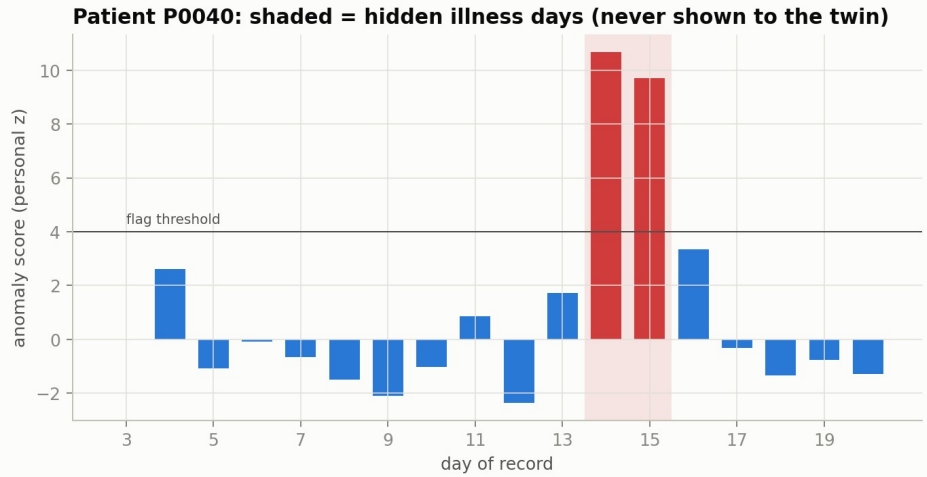
Correlation with true effect	Personalised twin	EHR-only twin	Population model
20-min walk after lunch	**0.68**	0.52	0.43
Rice → foxtail millet	**0.93**	0.89	0.89
30% smaller portion	**0.91**	0.85	0.86
Ranking patients by usual post-lunch peak	**0.92**	0.72	0.65

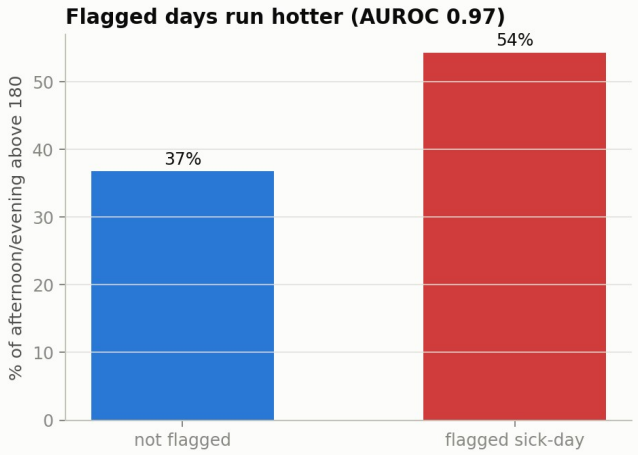
Why it matters

- Average peak drop: millet swap 53 mg/dL, smaller portion 40, walk 14
- But responses differ person to person; the personalised twin ranks who benefits most
- Stress test with different physics: rankings held, effect sizes biased
- Shown as simulated estimates for discussion, not prescriptions

A second warning: sick days from heart and sleep data

The brief's example of a weekly anomaly, made specific to diabetes care





0.97

AUROC for illness days, with no labelled training data

78%

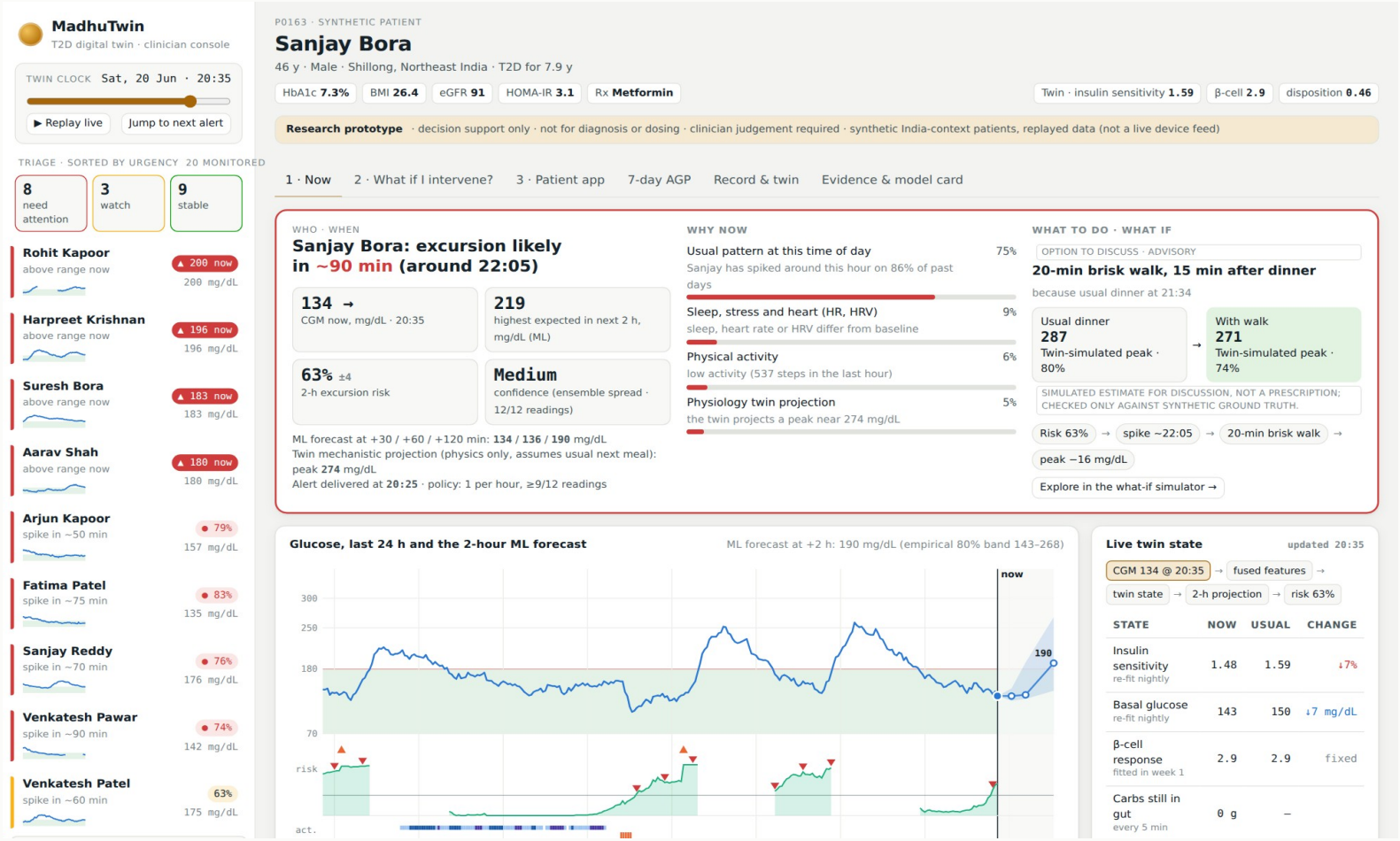
of illness days flagged; 0.12 false flags per patient-week

How it works and why it matters

- Every morning, last night's sleeping heart rate, HRV, sleep and yesterday's activity are compared with the patient's own previous 7 nights (personal z-scores)
- Flagged days ran hotter: 54% of the rest of the day above 180 mg/dL vs 37% otherwise
- Triggers 'sick-day rules': check glucose more often, fluids, review SGLT2 inhibitor / metformin if dehydrated, call the doctor
- Caveat: illness effects in the simulator are clear-cut; real-world infection detection from wearables is weaker but well documented

The doctor's console

Static web app: runs offline in any browser, no install

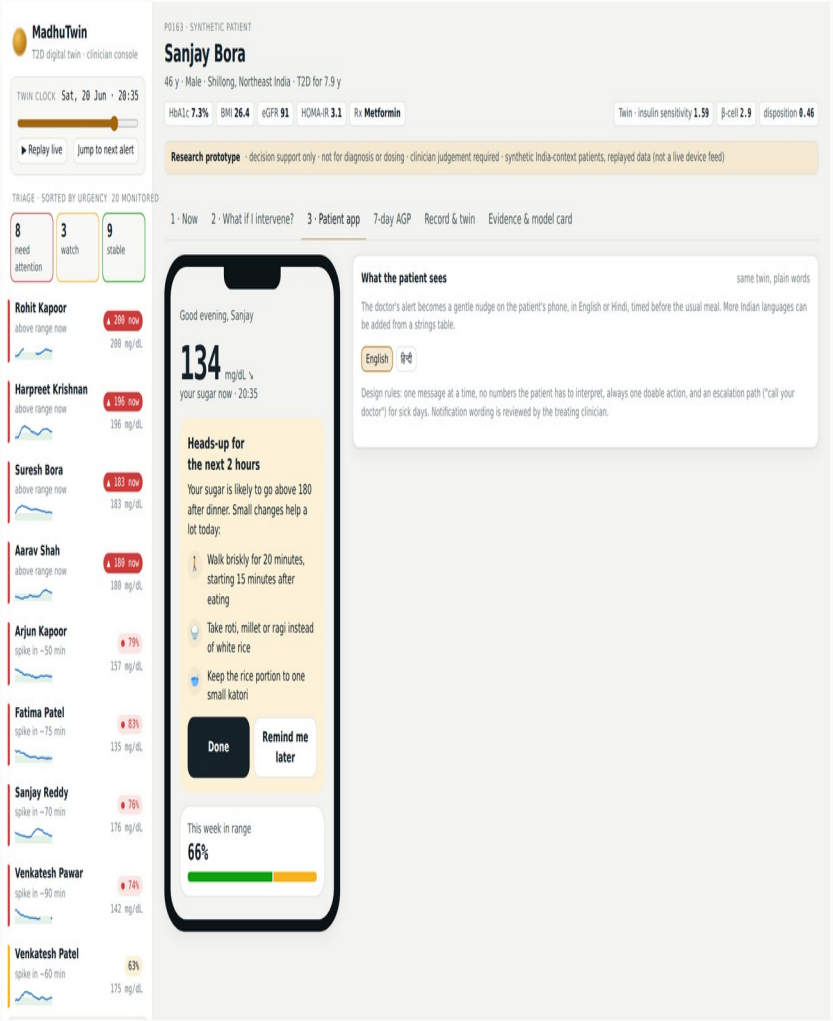


A doctor can

- Who: triage by risk × severity × confidence × time to event
- When: “spike likely in ~1 h 20 m, peak ~230 mg/dL”, with confidence
- Why: reasons in plain words, with % share
- What to do: one suggested action
- What if: simulate it → peak, time >180, event probability
- Twin state today vs the patient's own baseline

Designed with a clinic deployment pathway

Research prototype · interoperable outputs · offline-capable · patient-facing



Patient app nudges

- One message, one action, before the usual meal
- English and हिन्दी today; strings table for more languages
- Sick-day banner with 'call doctor'

FHIR R4 sample output

- Bundle: RiskAssessment + Observations
- Pseudonymous Patient (DPDP data minimisation)
- ABDM integration: proposed production path

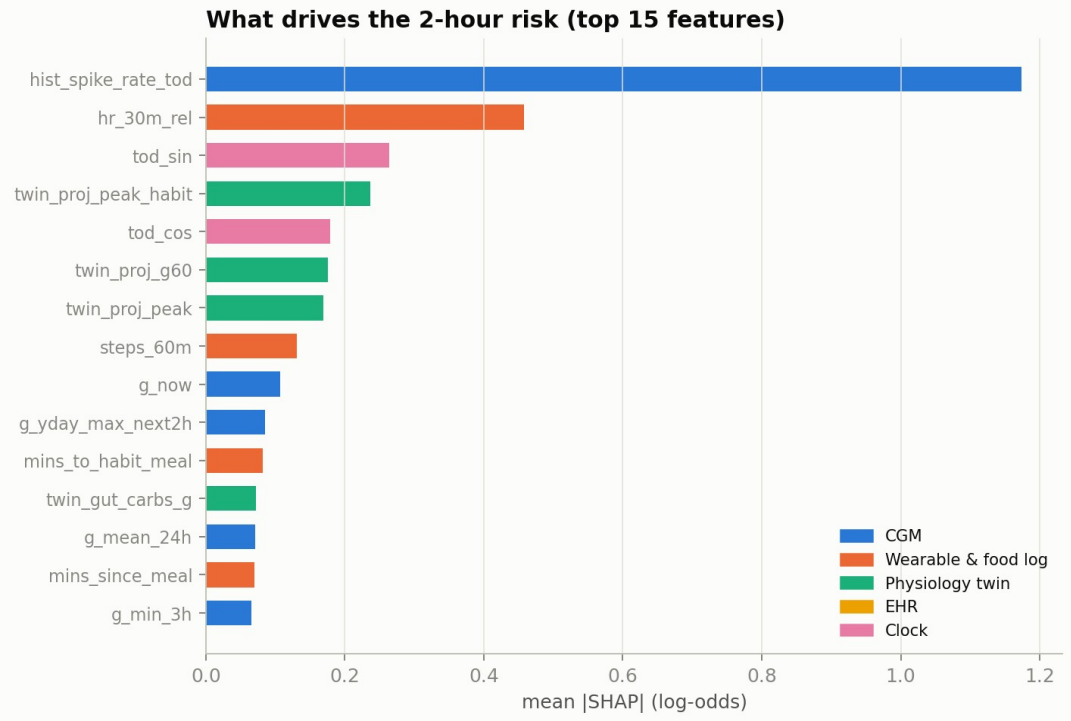
Twin as a service

- REST API (FastAPI): risk, forecast, what-if, FHIR
- Docker image serves API + console
- Runs on a laptop CPU; no GPU or cloud needed

Engineering quality

- 28 automated tests, none skipped: alert matching, API contract, leakage
- GitHub Actions CI
- One command: python run.py

Explainable, private, honest about limits



Top drivers of risk (SHAP attributions, not causes): the patient's own time-of-day spike history, heart rate relative to baseline, time of day, and the twin's projected peak including the usual next meal.

Privacy & compliance

- Synthetic patients + open de-identified cohorts; no PHI anywhere
- FHIR R4 sample output; ABDM integration is a proposed path
- Designed for on-premise inference; consent-first under the DPDP Act 2023
- Model card and decision-support labelling in the console

Limitations

- Main results are synthetic; open-cohort check: CGM + labs only, 10 with T2D; replayed data, no live devices
- Twin and simulator share equations: fidelity is an upper bound; what-if sizes biased under mismatch
- Twin adds little AUROC once 200+ patients exist
- Not a medical device; decision support only

From prototype to clinic

What we would build next

Now

300 synthetic patients; 265-person open-cohort check; nightly twin; console, patient app, FHIR sample, API, CI

Next 3 months

Validate on an Indian CGM cohort with meals and wearables; hypoglycaemia risk; shadow-mode run in one clinic

6 months

Live ABDM / Health Connect connectors; more Indian languages; clinician usability study

12 months

Prospective pilot with a diabetes clinic; weekly cardiovascular-risk twin from sleep and HRV; CDSCO SaMD pathway

Impact

The aim, still to be tested in a clinic: turn CGM from a retrospective report into a proactive, personalised coach, with advice tailored to how each patient responds and a screen doctors can act on in a two-minute consult.

Thank you

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